

DIPAM SEN

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ACADEMIC QUALIFICATIONS

Indian Institute of Technology, Kharagpur

Kharagpur, India

Bachelor of Technology, Computer Science and Engineering (CGPA: 9.86)

2024 – 2028

AWARDS AND ACHIEVEMENTS

- Secured **All India Rank 331** in **Joint Entrance Exam (Advanced) 2024** among 250,000 candidates.
- Secured **All India Rank 220** in **Joint Entrance Exam (Mains) 2024** among 1,200,000 candidates.
- Achieved the title of **Expert** on Codeforces with a **peak rating of 1632** (Handle: *dipamsen*)

PROJECTS

Lumen: Graphics Rendering DSL 🛠️ | OCaml, Raylib, FFmpeg | *Prof. Jibesh Patra* Feb'26 – Apr'26

- Developed a simple **statically-typed scripting language** for **graphics rendering** and **simulations**.
- Built a **compiler pipeline** in OCaml that transpiles Lumen scripts into C code using the **Raylib** library.
- Designed core language features including **functions**, **event blocks**, **vector support**, and **drawing primitives**.
- Implemented support for both **interactive graphics** applications and **video rendering** with FFmpeg integration.

Raft Distributed Consensus Engine 🛠️ | Go, Distributed Systems, Concurrency May'26 – Jun'26

- Implemented the Raft consensus algorithm, with **log replication** and **synchronization** across distributed nodes.
- Designed concurrent communication between cluster nodes using **goroutines** and **synchronization primitives**.
- Ensured **safety** and **consistency guarantees** under **node failures**, **network partitions**, and **leader election**.
- Built a replicated state machine with **persistent logs** and **majority-based commits** following the Raft protocol.

CPBattles 🛠️ | Node.js, React, MongoDB, PostgreSQL, CI/CD Pipelines Jun'25 – Jul'25

- Built a full stack web app with **Express** backend and **React** frontend for realtime programming battles.
- Integrated with **Codeforces API** to manage user verifications and problem submissions during battles.
- Managed scheduled tasks using **Agenda** with **MongoDB** and persistent data storage with **PostgreSQL**.
- Configured **CI/CD pipelines** to deploy the backend to Azure App Service.

IQPS: Intelligent QP Search 🛠️ | **metaKGP** | Rust Jun'25 – Present

- Maintainer of **IQPS**, a **Rust** web app serving 10K+ **previous year papers** to thousands of students.
- Integrated **Slack notifications** for paper uploads and imports, enabling real-time monitoring and updates.
- Streamlined **admin dashboard workflows** for **approving**, **editing**, **replacing** and **deleting** papers.
- Built a bulk paper **import pipeline** with **efficient deduplication** and **transactional rollback** on failure.

POSITIONS OF RESPONSIBILITY

- **Admin, metaKGP** 🛠️ Feb'26 – Present
 - Contributed to **multiple open source projects** maintained by **metaKGP** for the student community.
 - Organised events for familiarising the student community to open source and contributing to open source software.
- **Core Team Member, Kharagpur Open Source Society** Jul'25 – Present
 - Conducted **Kharagpur Winter of Code**, a program on open source with 360+ participants, 40+ projects.
 - Organized hands-on workshops and tutorials to familiarise students with Git/GitHub, Shell, and Linux.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, Rust, Go, C/C++, OCaml, Bash
- **Frameworks and Libraries:** React, Express, Node.js, Flask, Next.js, Axum, Tokio
- **Technologies:** Git/GitHub, Docker, Typst, LaTeX, MongoDB, PostgreSQL, GitHub Actions, Linux

COURSEWORK

Programming and Data Structures (Theory and Lab) | Algorithms (Theory and Lab) | Systems Programming Laboratory | Discrete Structures | Probability and Statistics | Signals and Systems | Formal Language and Automata Theory | Software Engineering (Theory and Lab) | Switching Circuits and Logic Design